

Retail Data Analysis using Tableau

To analyze retail sales data using Tableau and create an interactive dashboard for business decision-making.

❖ **Objectives**

- To understand sales performance across categories
- To analyze sales trends over time
- To identify profitable and loss-making areas
- To compare regional performance
- To build an interactive dashboard

❖ **Dataset Description**

The dataset contains retail order data (400+ records) with the following fields:

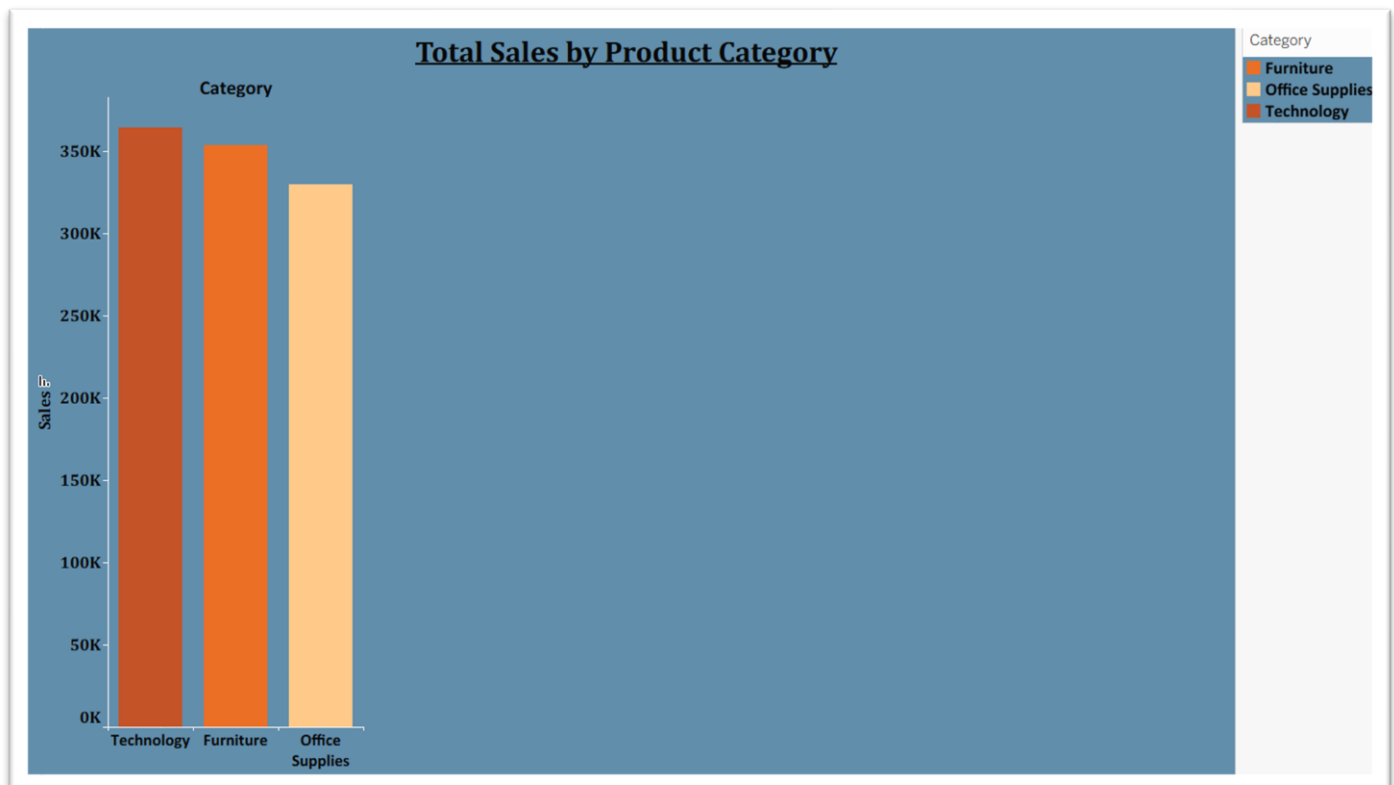
- Order Date
- Category & Sub-Category
- Region
- Sales
- Profit
- Quantity

❖ **Tools Used**

- Tableau Public
- Microsoft Excel

❖ **Tasks**

Task 1: Category Performance Analysis



- **Business Need:**

Management wants to identify which product category generates the highest revenue so that they can focus more on profitable areas.

- **Visualization Used:**

A Bar Chart is used to compare total sales across different product categories.

- **Observation:**

From the chart, it is clearly visible that Technology has the highest sales among all categories. It is followed by Furniture, while Office Supplies has comparatively lower sales.

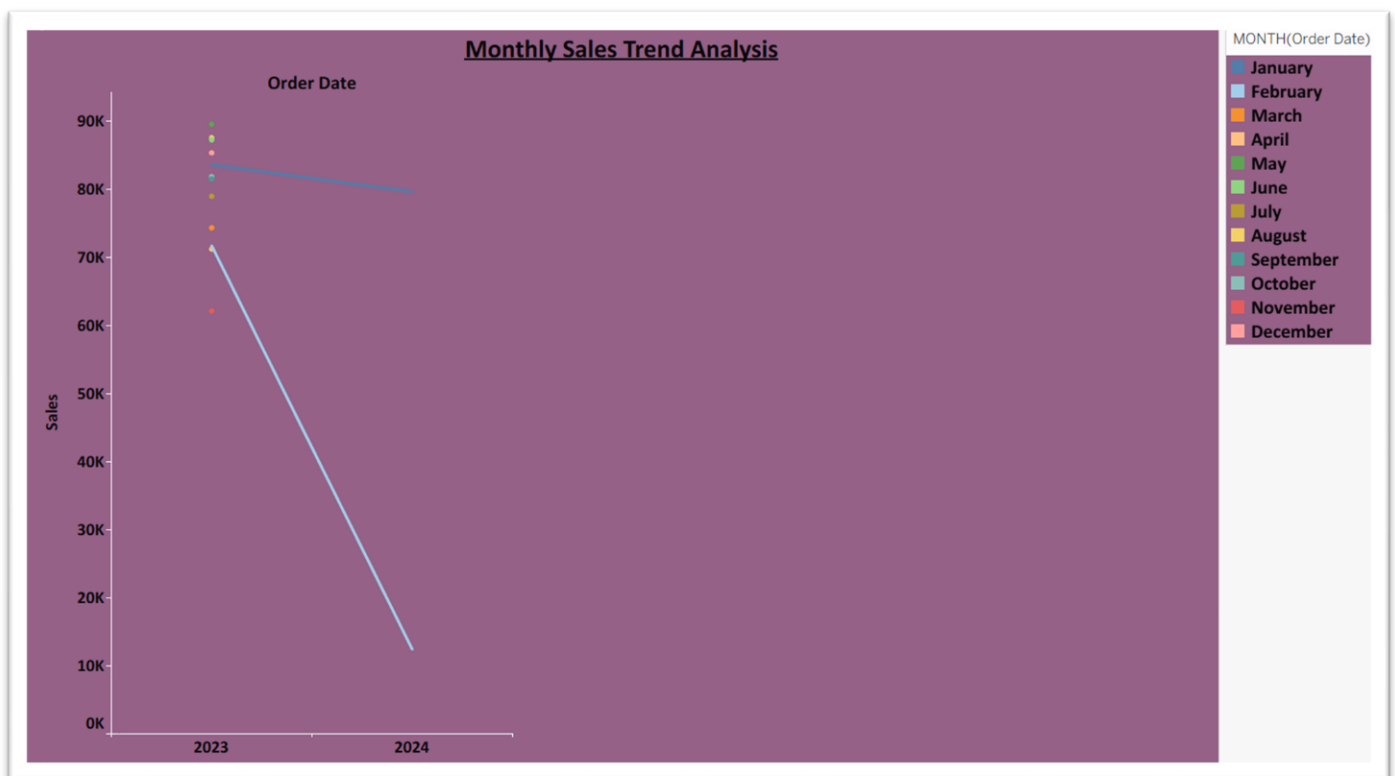
- **Conclusion**

The analysis shows that Technology products are the main revenue drivers for the business. This means customers are spending more on technology-related items compared to other categories. Furniture is also performing well, but Office Supplies is lagging behind.

So, the company should:

- Focus more on expanding the Technology category
- Maintain performance in Furniture
- Improve strategies for Office Supplies (like discounts or promotions)

Task 2: Sales Trend Analysis



- **Business Need:**

The leadership team wants to understand how sales change over time so they can plan promotions, manage inventory, and prepare for high or low demand periods.

- **Visualization Used:**

A Line Chart is used to analyze monthly sales trends over time.

- **Observation:**

The chart shows that sales generally increase over time, indicating business growth. There are some months where sales peak, while a few months show slower performance.

- **Conclusion**

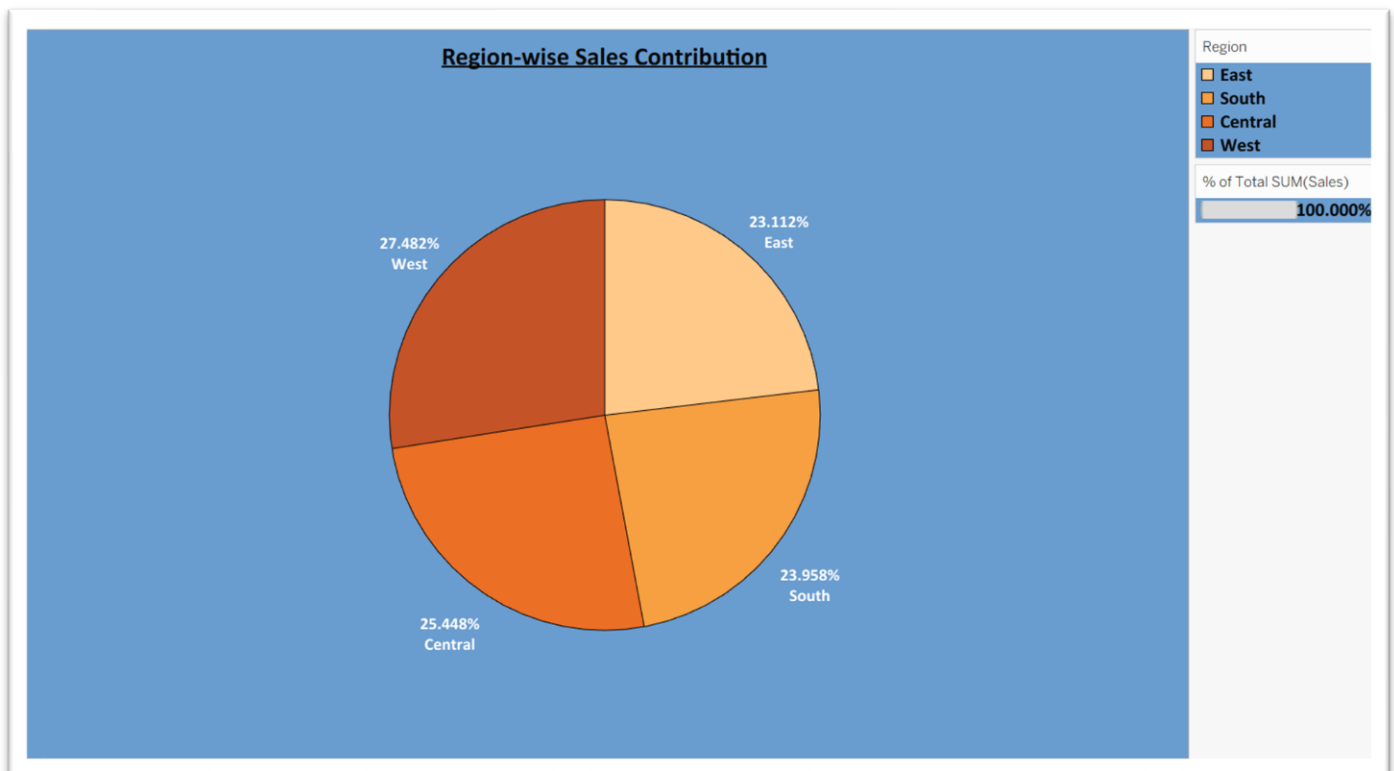
From the analysis, it is clear that sales do not remain constant throughout the year — they fluctuate. Some months perform better than others, which means demand changes over time.

The increasing trend also indicates that the business is growing steadily.

So, the company should:

- Plan promotions during slow months to boost sales
- Ensure sufficient inventory during peak months
- Focus on understanding seasonal demand patterns

Task 3: Regional Contribution Analysis



- **Business Need:**

Regional managers want to understand how much each region contributes to total sales so they can identify strong and weak markets.

- **Visualization Used:**

A Pie Chart is used to represent the percentage contribution of each region to total sales.

- **Observation:**

The chart shows that the West region contributes the highest share of total sales.

It is followed by Central and South regions, while the East region contributes the least.

- **Conclusion**

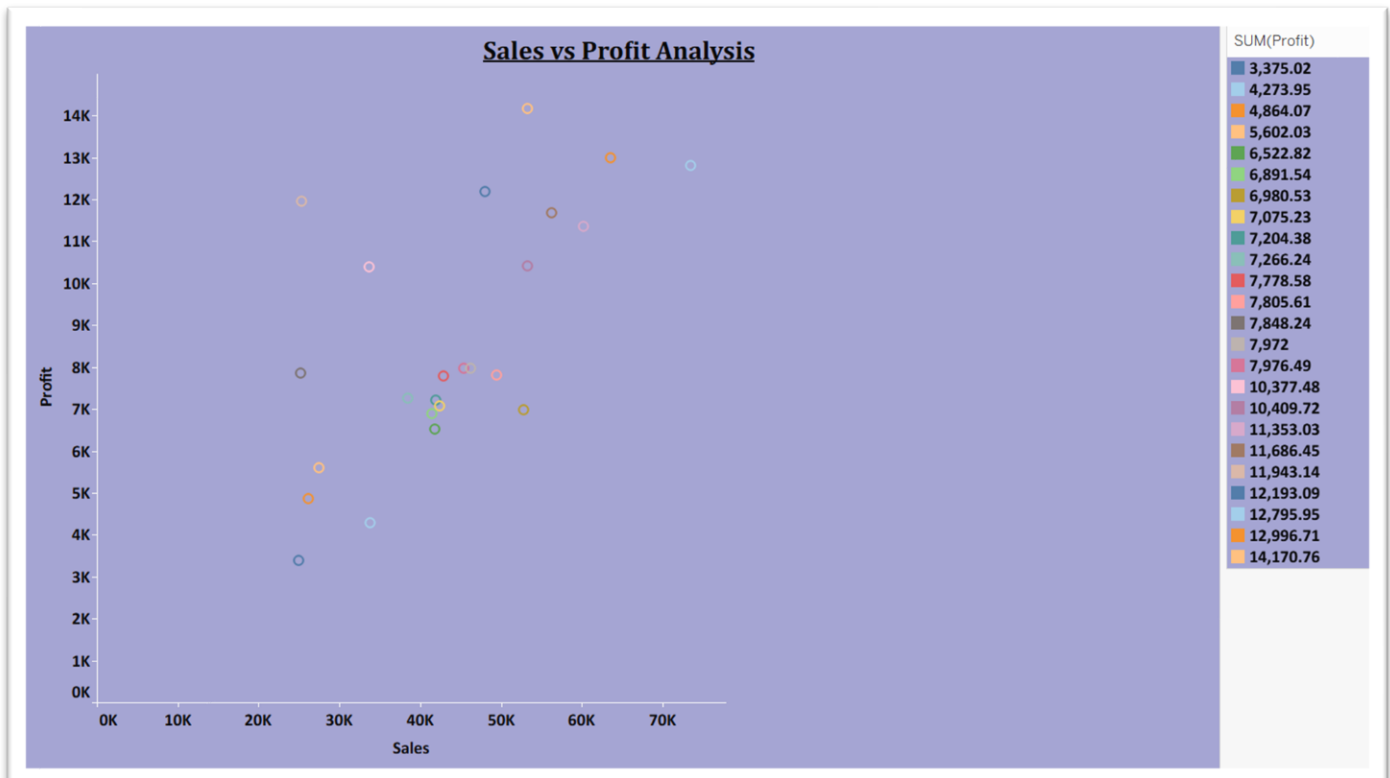
From the analysis, it is clear that not all regions perform equally. The West region is the strongest market and plays a major role in overall revenue.

On the other hand, regions like East are contributing less, which shows there is scope for improvement.

So, the company should:

- Focus on maintaining strong performance in the West region
 - Develop strategies (like marketing or offers) to improve sales in weaker regions
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Task 4: Profitability Investigation



- **Business Need:**

The finance team wants to check whether some high-sales orders are actually reducing overall profit, meaning sales are high but profit is low or even negative.

- **Visualization Used:**

A Scatter Plot is used to analyze the relationship between Sales and Profit.

- **Observation:**

From the chart, it can be seen that:

- Some data points show high sales with high profit (good performance)
- However, a few points show high sales but low or negative profit

- **Conclusion**

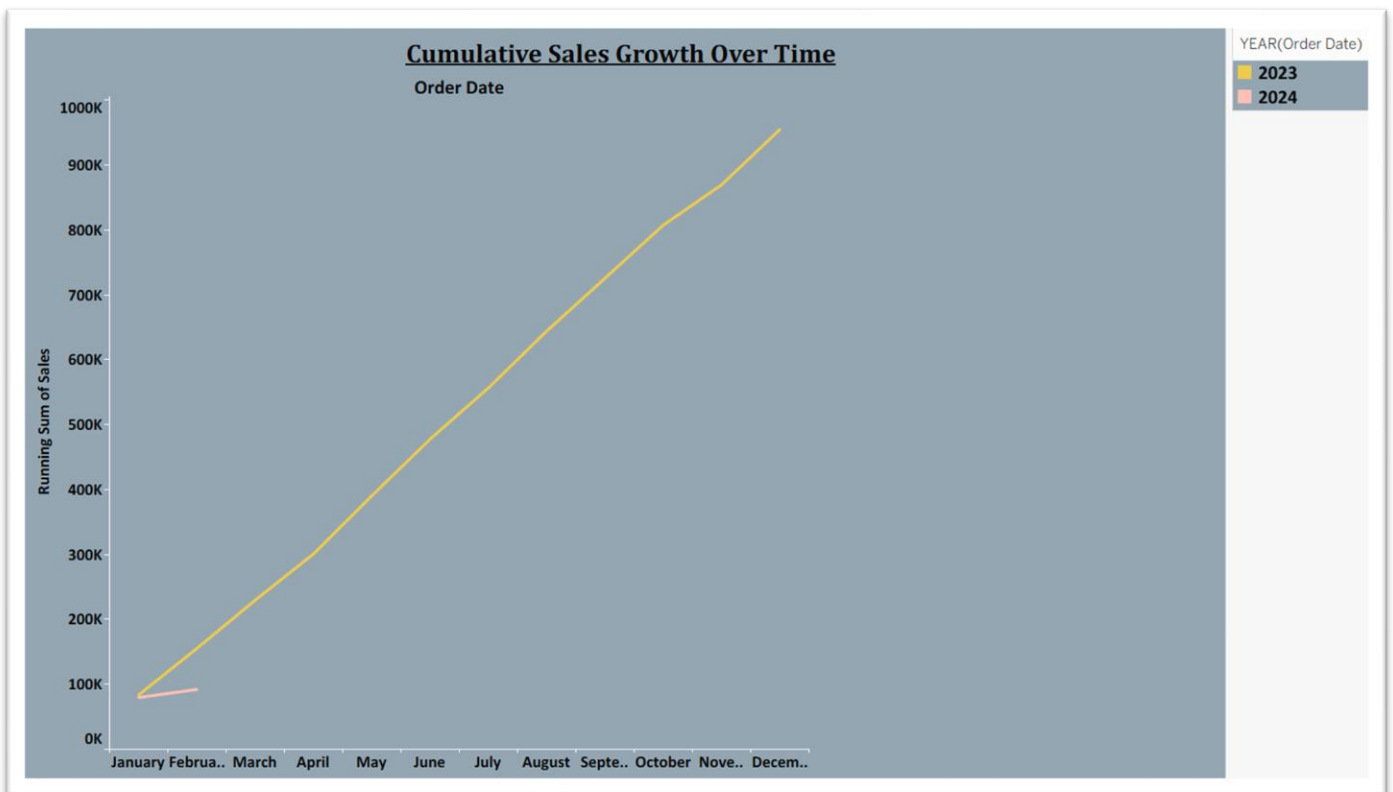
The analysis clearly shows that higher sales do not always mean higher profit. Some orders generate good revenue but still result in low profit or even losses, possibly due to high discounts, costs, or inefficient pricing.

This means the company should not focus only on increasing sales but also on maintaining profitability.

So, the company should:

- Identify and reduce loss-making products or orders
- Review pricing and discount strategies
- Focus on products that give both high sales and high profit

Task 5: Business Growth Evaluation



- **Business Need:**

Senior management wants to understand whether the business is growing consistently over time so they can evaluate overall performance and future planning.

- **Visualization Used:**

A Cumulative Line Chart is used to show the running total of sales over time.

- **Observation:**

The chart shows a continuous upward trend in cumulative sales.

There are no major drops, and sales keep increasing month by month.

- **Conclusion**

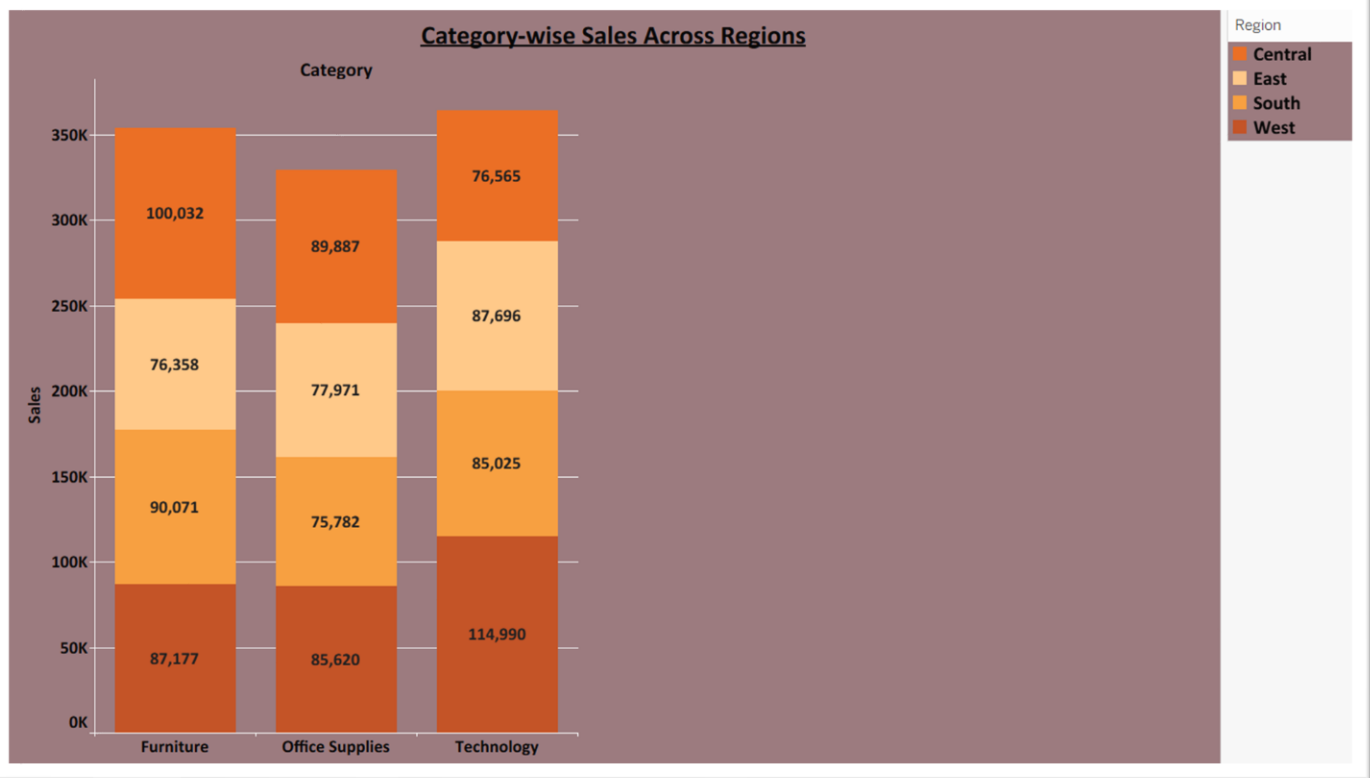
From the analysis, it is clear that the business is growing steadily over time. The continuous rise in cumulative sales indicates that the company is consistently generating revenue and expanding its performance.

This is a positive sign for the business, showing stability and growth.

So, the company should:

- Continue its current strategies that are driving growth
 - Focus on sustaining this growth in the long term
 - Identify opportunities to further accelerate sales
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Task 6: Category vs Region Performance



- **Business Need:**

The operations team wants to understand which regions perform better for each product category so they can focus on strong markets and improve weaker ones.

- **Visualization Used:**

A Stacked Bar Chart is used to compare category-wise sales across different regions.

- **Observation:**

The chart shows that:

- Each category (Technology, Furniture, Office Supplies) receives contributions from all regions
- However, some regions contribute more in specific categories
- Technology category shows the highest overall sales across regions

- **Conclusion**

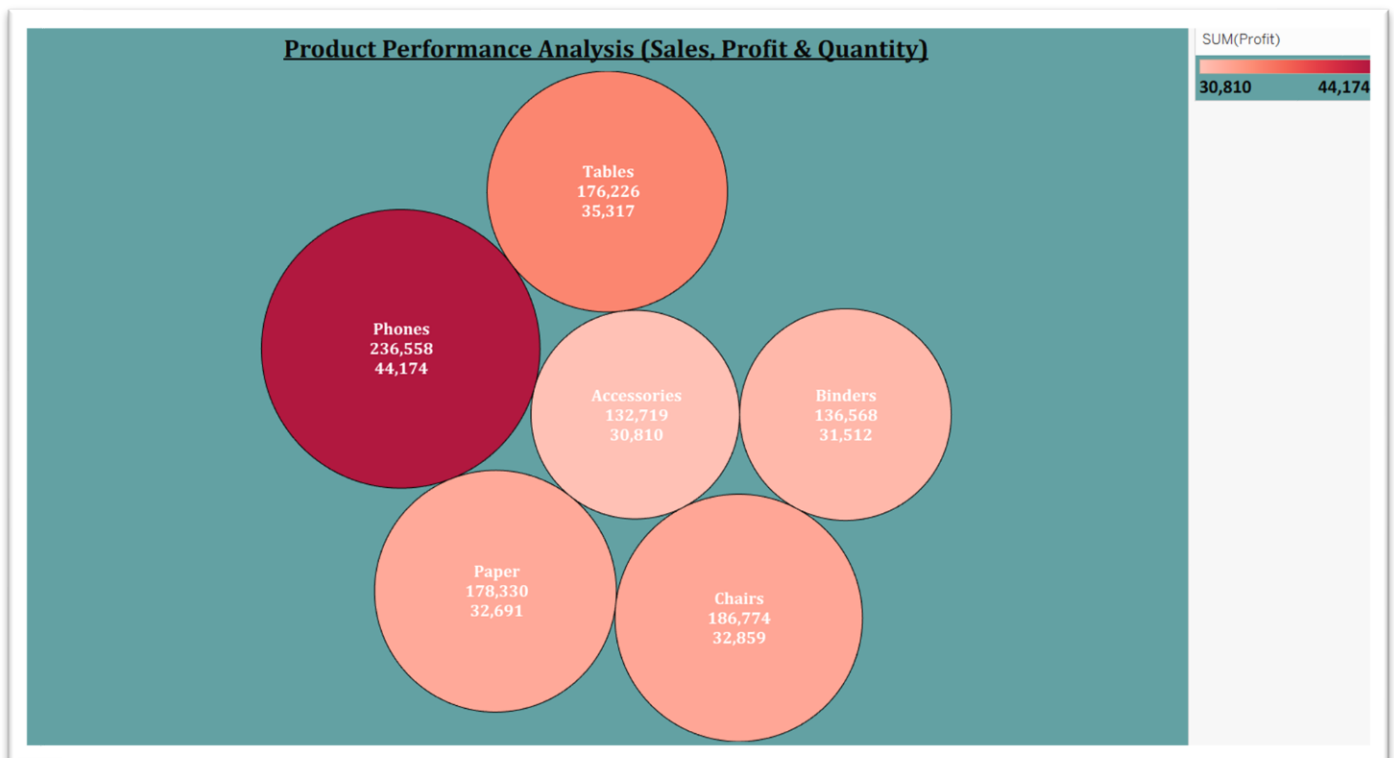
From the analysis, it is clear that sales performance varies by both category and region. Some regions are stronger in certain product categories, which means customer demand is not the same everywhere.

This helps the company understand where each product category is performing well and where it needs improvement.

So, the company should:

- Focus on strong regions for each category to maximize sales
- Improve performance in weaker regions using targeted strategies
- Customize marketing and inventory based on regional demand

Task 7: Product Prioritization



- **Business Need:**

The product team wants to identify which products have the highest impact on the business so they can focus on the most valuable ones.

- **Visualization Used:**

A Bubble Chart is used to analyze multiple factors together:

- Sales (size of bubble)
- Profit (position)
- Quantity (additional dimension)

- **Observation:**

The chart shows that:

- Some products (like Phones and Chairs) have high sales and high profit
- These products also have good quantity sold
- A few products show low profit despite decent sales

- **Conclusion**

From the analysis, it is clear that not all products contribute equally to the business. Some products generate high sales, profit, and quantity, making them high-priority products.

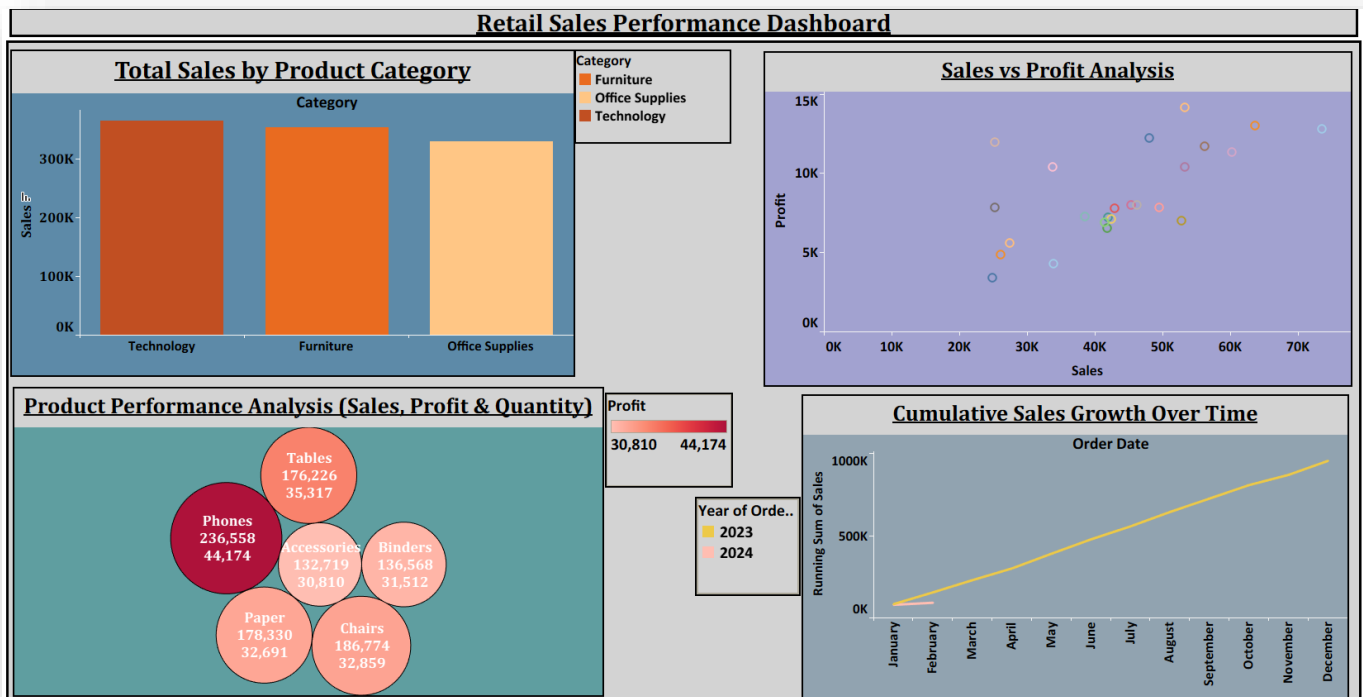
Products like Phones and Chairs are key drivers of business success and should be given more attention.

On the other hand, some products may sell well but generate low profit, so they need improvement.

So, the company should:

- Focus on high-performing products to maximize growth
- Improve or review low-profit products
- Plan inventory and marketing based on product performance

Task 8: Dashboard



- **Business Need:**

The CEO wants a single, clear view of the entire business performance so that important decisions can be taken quickly without going through multiple reports.

- **Visualization Used:**

An Interactive Dashboard is created by combining multiple charts such as:

- Category Performance (Bar Chart)
- Sales Trend (Line Chart)
- Sales vs Profit (Scatter Plot)
- Product Performance (Bubble Chart)

Filters like Region and Category are applied to make the dashboard interactive.

- **Observation:**

The dashboard provides a complete overview of sales, profit, growth, and product performance in one place.

Users can easily filter data based on region or category to get specific insights.

- **Conclusion**

The dashboard successfully brings together all important business information in a simple and interactive format. Instead of checking multiple reports, the CEO can now quickly understand performance, identify problems, and make decisions.

The use of filters makes it flexible, allowing users to focus on specific regions or categories as needed.

So, the dashboard helps in:

- Quick decision-making
- Better understanding of business performance
- Identifying growth opportunities and problem areas

Q9. Chart Selection Justification

Explain why different charts were used for different business questions in this analysis.

- **Explanation:**

Different types of charts are used in this analysis because each chart is suitable for answering a specific business question in the most clear and effective way.

- **Justification**

- Bar Chart (Category Performance):

Bar chart is used because it makes it very easy to compare sales across different categories. It clearly shows which category is highest and lowest.

- Line Chart (Sales Trend & Growth):

Line chart is best for showing changes over time. It helps in understanding trends, growth patterns, and identifying peak and slow periods.

- Pie Chart (Regional Contribution):

Pie chart is used to show percentage contribution. It helps in quickly understanding how much each region contributes to total sales.

- Scatter Plot (Sales vs Profit):

Scatter plot is useful to show the relationship between two variables (Sales and Profit). It helps in identifying loss-making or low-profit orders.

- Stacked Bar Chart (Category vs Region):

This chart is used to compare multiple categories across different regions in one view. It helps in identifying strong and weak market areas.

- Bubble Chart (Product Performance):

Bubble chart allows multi-dimensional analysis (Sales, Profit, Quantity) in a single view, making it easy to identify high-impact products.

- **Conclusion:**

Each chart is selected based on its ability to present data clearly and answer a specific business question effectively, which improves understanding and decision-making.

Q10. Business Insight Summary

Summarize two key insights from your dashboard and explain how they can help management make better decisions.

- **Insight 1: Technology is the Top-Performing Category**

Explanation:

From the dashboard, it is clear that the Technology category generates the highest sales compared to other categories.

How it helps management:

This insight helps management understand where the majority of revenue is coming from. They can:

- Invest more in Technology products
- Increase stock and variety
- Focus marketing efforts on this category

This will help in maximizing revenue and business growth.

- **Insight 2: High Sales Do Not Always Mean High Profit**

Explanation:

The Sales vs Profit analysis shows that some high-sales orders have low or even negative profit.

How it helps management:

This insight highlights that simply increasing sales is not enough.

Management should:

- Review pricing and discount strategies
- Identify loss-making products
- Focus on improving profit margins

This will help in increasing overall profitability, not just sales.

- **Conclusion**

These insights show that the business needs to focus on both high-performing categories and profitability control. By using these findings, management can take smarter, data-driven decisions.
